

Task 6 - Data Transformation & Feature Engineering

Introduction

Data transformation and feature engineering are important steps in data preprocessing. These techniques improve data quality and help machine learning models achieve better performance.

Objective:

The objective of this task is to transform the California Housing Dataset using feature scaling and create new features that can provide additional information for predictive modeling.

Tools Used:

- Python
- Pandas
- Scikit-learn
- Google Colab

Dataset:

California Housing Dataset

The dataset contains information about housing characteristics such as median income, house age, average rooms, population, latitude, longitude, and median house value.

Steps Performed:

1. Data Loading

The California Housing Dataset was loaded using the Scikit-learn library and converted into a Pandas DataFrame.

2. Data Exploration

The dataset structure was analyzed using:

- `df.info()`
- `df.describe()`

This helped in understanding the number of records, data types, and statistical summary of the dataset.

3. Feature Scaling

`StandardScaler` was applied to standardize all numerical features.

Benefits of Scaling:

- Improves model performance.
- Prevents features with large values from dominating the model.
- Helps optimization algorithms converge faster.

4. Feature Engineering

A new interaction feature was created:

`Income_Rooms = MedInc × AveRooms`

This feature combines median income and average rooms to capture their relationship and provide additional information for machine learning models.

5. Saving the Dataset

The transformed dataset was exported as:
engineered_housing_dataset.csv

Results:

- Successfully loaded and explored the dataset.
- Applied feature scaling using StandardScaler.
- Created a new interaction feature named Income_Rooms.
- Generated an engineered dataset ready for machine learning applications.

Interview Questions

Q1. Scaling vs Norm

Scaling (Standardization):

- Mean becomes 0.
- Standard deviation becomes 1.

Normalization:

- Values are transformed between 0 and 1.

Q2. Why use One-Hot Encoding?

One-Hot Encoding converts categorical variables into numerical format. Machine learning algorithms cannot directly process text categories, so One-Hot Encoding creates separate binary columns for each category.

Conclusion

Data transformation and feature engineering were successfully performed on the California Housing Dataset. The resulting dataset is more suitable for machine learning models and can help improve prediction accuracy.